```
➜  docker-py-java git:(main) ✗ mkdir py-task-1
➜  docker-py-java git:(main) ✗ cd py-task-1
➜  py-task-1 git:(main) ✗ cat > app.py << 'EOF'
heredoc> import random
heredoc> num = [random.randint(1,100) for i in range(10)]
heredoc> print("Random list of 10 numbers: ",num)
heredoc> EOF
➜  py-task-1 git:(main) ✗ cat > Dockerfile << 'EOF'

heredoc> FROM python:3.10-slim
heredoc> WORKDIR /app
heredoc> COPY app.py .
heredoc> CMD ["python","app.py"]
heredoc> EOF
```

```
➜  py-task-1 git:(main) ✗ docker build -t python-task1 .
[+] Building 26.7s (9/9) FINISHED                                                
```

```
➜  py-task-3 git:(main) ✗ docker build -t python-task-3 .
[+] Building 2.1s (9/9) FINISHED                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                              0.0s
 => => transferring dockerfile: 110B                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.10-slim               2.0s
 => [auth] library/python:pull token for registry-1.docker.io                     0.0s
 => [internal] load .dockerignore                                                 0.0s
 => => transferring context: 2B                                                   0.0s
 => [1/3] FROM docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b46  0.0s
 => => resolve docker.io/library/python:3.10-slim@sha256:6ff506466f8b1e981719b46  0.0s
 => [internal] load build context                                                 0.0s
 => => transferring context: 480B                                                 0.0s
 => CACHED [2/3] WORKDIR /app                                                     0.0s
 => [3/3] COPY app.py .                                                           0.0s
 => exporting to image                                                            0.0s
 => => exporting layers                                                           0.0s
 => => exporting manifest sha256:0c49118e3e58aee690aa2f1928dfcb130ad1c93916e9d5e7  0.0s
 => => exporting config sha256:779956d55da1c6e5dce3ff439548b139febb295abbfeb081c  0.0s
 => => exporting attestation manifest sha256:27d21ad8f84f29d1b449181a6942e64473c4  0.0s
 => => exporting manifest list sha256:d12e4a552ff1d83d6abe21f76cb53728817e5d65f22  0.0s
 => => naming to docker.io/library/python-task-3:latest                           0.0s
 => => unpacking to docker.io/library/python-task-3:latest                        0.0s
```

```
➜  py-task-3 git:(main) ✗ docker run --rm -it python-task-3
Enter an actor's name: prabhas
Funny dialogue: Before saving the kingdom, let me finish my biryani!
```

2

```
➜  docker-py-java git:(main) ✗ mkdir java-t-1
➜  docker-py-java git:(main) ✗ cd java-t-1
```

```
➜  java-t-1 git:(main) ✗ docker build -t j-t-1 .
[+] Building 13.4s (10/10) FINISHED                               docker:desktop-linux
 => [internal] load build definition from Dockerfile                              0.0s
 => => transferring dockerfile: 132B                                              0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk         3.1s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io            0.0s
 => [internal] load .dockerignore                                                 0.0s
 => => transferring context: 2B                                                   0.0s
 => [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d61  9.6s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d61  0.0s
 => => sha256:97ea7a078a6c8f4aaeb8fa58a0658f6d3631d002252fc264f6be73a87eee7eb8 2.  0.4s
 => => sha256:12a4be5f67a4d7dfd447bc299b563bdc9b95b1f453c176a1332ebb96f75a9e21 14  6.8s
 => => sha256:70cb2bb6cf6f828366577597e6891d2194b4084dd90082d604d1673f1b6cb575 158  0.6s
 => => sha256:787458d0162650daabd0e631a9c7a9b07e87bbe6c06ee05c4777995a1121f102 27  4.7s
 => => sha256:5fa77813dc560f299656013d742628128cac51fdfda979349e96a9f09497170f 392  0.5s
 => => sha256:06db5ba23986104cd78dcfa853ddf2354c943d2efefc333eca2b447d4559d89c 40  7.4s
 => => extracting sha256:06db5ba23986104cd78dcfa853ddf2354c943d2efefc333eca2b447  0.5s
 => => extracting sha256:5fa77813dc560f299656013d742628128cac51fdfda979349e96a9f  0.0s
 => => extracting sha256:787458d0162650daabd0e631a9c7a9b07e87bbe6c06ee05c4777995  0.3s
 => => extracting sha256:12a4be5f67a4d7dfd447bc299b563bdc9b95b1f453c176a1332ebb96  0.7s
 => => extracting sha256:70cb2bb6cf6f828366577597e6891d2194b4084dd90082d604d1673  0.0s
 => => extracting sha256:97ea7a078a6c8f4aaeb8fa58a0658f6d3631d002252fc264f6be73a  0.0s
 => [internal] load build context                                                 0.0s
 => => transferring context: 589B                                                 0.0s
 => [2/4] WORKDIR /app                                                            0.3s
 => [3/4] COPY App.java .                                                         0.0s
 => [4/4] RUN javac App.java                                                      0.3s
```

```
➜  java-t-1 git:(main) ✗ docker run --rm -it j-t-1
Enter first number: 132
Enter second number: 432
Addition: 564
Multiplication: 57024
```

```
➜  docker-py-java git:(main) ✗ mkdir java-t-2
➜  docker-py-java git:(main) ✗ cd java-t-2
➜  java-t-2 git:(main) ✗ docker build -t j-t-2 .
[+] Building 2.4s (10/10) FINISHED                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                              0.0s
 => => transferring dockerfile: 132B                                              0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk         2.0s
 => [auth] library/eclipse-temurin:pull token for registry-1.docker.io            0.0s
 => [internal] load .dockerignore                                                 0.0s
 => => transferring context: 2B                                                   0.0s
 => [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d61  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d61  0.0s
 => [internal] load build context                                                 0.0s
 => => transferring context: 364B                                                 0.0s
 => CACHED [2/4] WORKDIR /app                                                     0.0s
 => [3/4] COPY App.java .                                                         0.0s
 => [4/4] RUN javac App.java                                                      0.3s
 => exporting to image                                                            0.0s
 => => exporting layers                                                           0.0s
 => => exporting manifest sha256:3313d128db352398cc1400467b66318d94fc2370efc32bf  0.0s
 => => exporting config sha256:6d04964b8558b6278db426fcd61b8e6e1814ab9c0edde0dd6  0.0s
 => => exporting attestation manifest sha256:86f3316cd31da55b3183c8f7ded98279a01  0.0s
 => => exporting manifest list sha256:125b49d2ea542444ca199bee01f235110febc17569  0.0s
 => => naming to docker.io/library/j-t-2:latest                                   0.0s
 => => unpacking to docker.io/library/j-t-2:latest                                0.0s
```

```
➜  java-t-2 git:(main) ✗ docker run --rm j-t-2
Animal eats food.
Dog barks.
```

```
→  docker-py-java git:(main) ✗ mkdir j-t-3
→  docker-py-java git:(main) ✗ cd j-t-3
→  j-t-3 git:(main) ✗ docker build -t j-t-3 .
[+] Building 2.9s (9/9) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                           0.0s
 => => transferring dockerfile: 132B                                                           0.0s
 => [internal] load metadata for docker.io/library/eclipse-temurin:17-jdk                      2.5s
 => [internal] load .dockerignore                                                              0.0s
 => => transferring context: 2B                                                                0.0s
 => [1/4] FROM docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d614e4e421f52e3211fc5aeaddd  0.0s
 => => resolve docker.io/library/eclipse-temurin:17-jdk@sha256:5d6042fb8cdc14d614e4e421f52e3211fc5aeaddd  0.0s
 => [internal] load build context                                                              0.0s
 => => transferring context: 811B                                                              0.0s
 => CACHED [2/4] WORKDIR /app                                                                  0.0s
 => [3/4] COPY App.java .                                                                      0.0s
 => [4/4] RUN javac App.java                                                                   0.3s
 => exporting to image                                                                         0.0s
 => => exporting layers                                                                        0.0s
 => => exporting manifest sha256:1e1913576a9c9e3de1341c51b56e5d8c431fb2b0cfb254b7185a370eda6eb7b3  0.0s
 => => exporting config sha256:b15dee0a828a892cfa9275d48cfb788bfe5b353d3527bd30e9c9ae580fe5f7e2  0.0s
 => => exporting attestation manifest sha256:0fec36a122ef9a9417046435c7f4ae3c018d1102178abbfc213cd08d22b  0.0s
 => => exporting manifest list sha256:a063c3b68c9c7d8d2a6c8928b297173bcdf4eb240158683a7f330069c1b257c8  0.0s
 => => naming to docker.io/library/j-t-3:latest                                                0.0s
 => => unpacking to docker.io/library/j-t-3:latest                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/z016i99p16q6u6dnv9ritjj3x
```

```
 git:(main) ✗ docker run --rm -it j-t-3
Enter a word or number: racecar
Reversed input: racecar
racecar is a palindrome.
→  j-t-3 git:(main) ✗ docker run --rm -it j-t-3
Enter a word or number: tim
Reversed input: mit
tim is not a palindrome.
```

```
→  7. DevOpsL git:(main) ✗ mkdir -p node-docker/node-task1

→  7. DevOpsL git:(main) ✗ cd node-docker/node-task1
```

```
→  node-task1 git:(main) ✗ docker build -t node-t1 .
[+] Building 8.4s (9/9) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                           0.0s
 => => transferring dockerfile: 104B                                                           0.0s
 => [internal] load metadata for docker.io/library/node:24-slim                                3.7s
 => [auth] library/node:pull token for registry-1.docker.io                                    0.0s
 => [internal] load .dockerignore                                                              0.0s
 => => transferring context: 2B                                                                0.0s
 => [1/3] FROM docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f047b5def2f02ea1dbbc2ca920407d46d7ec7f  4.4s
 => => resolve docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f047b5def2f02ea1dbbc2ca920407d46d7ec7f  0.0s
 => => sha256:49ac4e8e4295c8ee475ab0dff5e76277d9328670ecc925439122c1ba3f90b228 3.31kB / 3.31kB  0.8s
 => => sha256:4fafc9a89135b87d66ccaf8587cd7e44d65e717d06fb239c6fd4839e64678987 446B / 446B      0.3s
 => => sha256:9e7afa9c9ea06b755c905f765b2d70c431dd5c00e88bc0810ab04b5ab49b69c5 50.84MB / 50.84MB  2.5s
 => => sha256:c1a924fd4d24bc36aeda3ab83dc77c1e84a9c624e42b08ed33e9985744558059 1.71MB / 1.71MB  0.6s
 => => sha256:c75f989a229d12b2d2613a5997de9ff3546f664c22da9248720033a2410220f6 28.14MB / 28.14MB  3.2s
```

```
→  node-task1 git:(main) ✗ docker run --rm node-t1
Praise the Lord
```

```
→  node-docker git:(main) ✗ mkdir node-task2
→  node-docker git:(main) ✗ cd node-task2
```

```
→  node-task2 git:(main) ✗ docker build -t node-t2 .
[+] Building 2.0s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 104B
 => [internal] load metadata for docker.io/library/node:24-slim
 => [auth] library/node:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/node:24-slim@sha256:d6aa754f16b3
 => => resolve docker.io/library/node:24-slim@sha256:d6aa754f16b3
 => [internal] load build context
 => => transferring context: 263B
 => CACHED [2/3] WORKDIR /app
 => [3/3] COPY app.js .
 => exporting to image
 => => exporting layers
```

```
→  node-task2 git:(main) ✗ docker run --rm node-t2
Student Details
Name: Konala
Roll Number: 18261A05R0
Branch: Paper Engineering
```



```
→  node-docker git:(main) ✗ mkdir node-task3
→  node-docker git:(main) ✗ cd node-task3
→  node-task3 git:(main) ✗ docker build -t node-task3 .
[+] Building 2.1s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 117B
 => [internal] load metadata for docker.io/library/node:24-slim
 => [auth] library/node:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f047b5def2
 => => resolve docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f047b5def2
 => [internal] load build context
 => => transferring context: 351B
 => CACHED [2/3] WORKDIR /app
 => [3/3] COPY app.js .
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:3cb890b7f9448cb1ed2db905dbc51b6f297ac33ac06e02c9829
 => => exporting config sha256:9a2388b495e041b2748f84d4feb77993f65bb52e5b1ef63b15fc4
 => => exporting attestation manifest sha256:95b3604869f3a8cca3cf2adbabd19babd03b4ac
 => => exporting manifest list sha256:887a856c358bbe332dcc5de48adab4788ad041fd5f13fc
 => => naming to docker.io/library/node-task3:latest
 => => unpacking to docker.io/library/node-task3:latest
```

```
→  node-task3 git:(main) ✗ docker run --rm -p 127.0.0.1:3000:3000 node-task3
Server listening on port 3000
```




```
→  node-task3 git:(main) ✗ cd ..
→  node-docker git:(main) ✗ mkdir node-task4
→  node-docker git:(main) ✗ cd node-task4
```

```
➜  node-task4 git:(main) ✗ docker build -t node-task4 .
[+] Building 1.9s (9/9) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 107B
 => [internal] load metadata for docker.io/library/node:24-slim
 => [auth] library/node:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f
 => => resolve docker.io/library/node:24-slim@sha256:d6aa754f16b3197301076f
 => [internal] load build context
 => => transferring context: 248B
 => CACHED [2/3] WORKDIR /app
 => [3/3] COPY app.js .
```

```
➜  node-task4 git:(main) ✗ docker run --rm node-task4
Current date and time:
Friday, 9 October 2026 at 2:38:09 pm IST
```

5

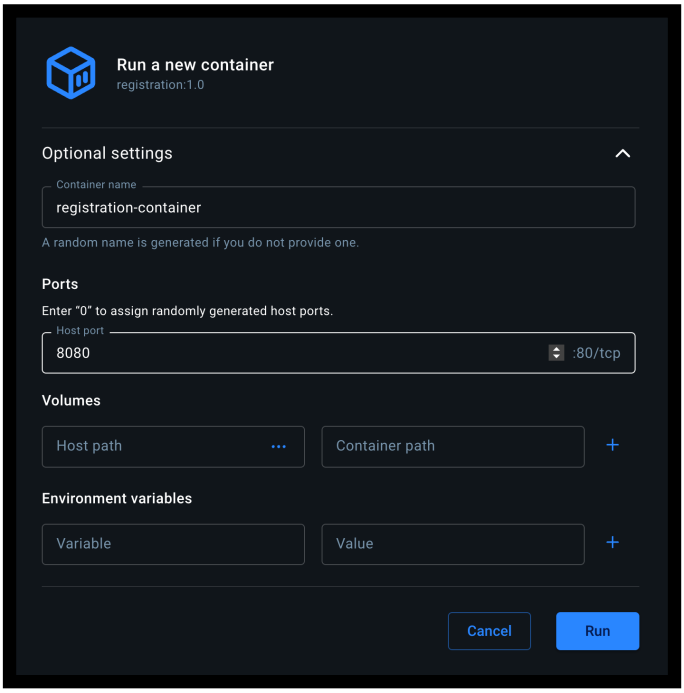
```
➜  node-task4 git:(main) ✗ cd ..
➜  node-docker git:(main) ✗ mkdir -p python-docker-compose-app

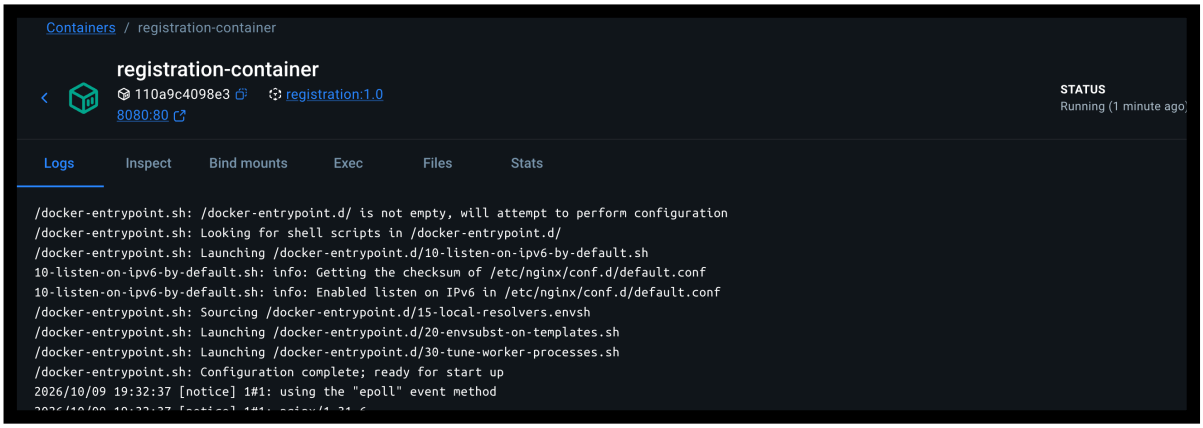
➜  node-docker git:(main) ✗ cd python-docker-compose-app
```

```
➜  python-docker-compose-app git:(main) ✗ docker compose config
name: python-docker-compose-app
services:
  guess_num:
    build:
      context: /Users/timothysaxena/Downloads/MGIT/Sem-V/7. DevOpsL/node-docker/python-docker-compose-app
      dockerfile: Dockerfile
    command:
      - python
      - -u
      - guess_num.py
    container_name: compose_guess_num_container
    networks:
      default: null
  ran_num:
    build:
      context: /Users/timothysaxena/Downloads/MGIT/Sem-V/7. DevOpsL/node-docker/python-docker-compose-app
      dockerfile: Dockerfile
    command:
      - python
      - -u
      - ran_num.py
    container_name: compose_ran_num_container
    networks:
      default: null
networks:
  default:
    name: python-docker-compose-app_default
```

```
➜  python-docker-compose-app git:(main) ✗ docker compose up --build
[+] Building 2.3s (13/13) FINISHED
[+] up 5/5
 ✔ Image python-docker-compose-app-guess_num Built
 ✔ Image python-docker-compose-app-ran_num   Built
 ✔ Network python-docker-compose-app_default Created
 ✔ Container compose_guess_num_container     Created
 ✔ Container compose_ran_num_container       Created
Attaching to compose_guess_num_container, compose_ran_num_container
compose_guess_num_container  | Generated guess: 44
compose_guess_num_container  | Wrong! The number was 24
compose_ran_num_container    | Numbers: [77, 82, 64, 22, 22, 41, 72, 17, 61, 64]
compose_ran_num_container    | Maximum: 82
compose_ran_num_container    | Minimum: 17
compose_ran_num_container    | Average: 52.2
compose_guess_num_container exited with code 0
compose_ran_num_container exited with code 0
```

```
→  registration-app git:(main) ✗ docker build -t registration:1.0 .
[+] Building 0.5s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 182B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/3] FROM docker.io/library/nginx:alpine@sha256:df221db836e175408919
 => => resolve docker.io/library/nginx:alpine@sha256:df221db836e175408919
 => [internal] load build context
 => => transferring context: 100B
 => CACHED [2/3] COPY src/registration.html /usr/share/nginx/html/index.h
 => CACHED [3/3] COPY src/login.html /usr/share/nginx/html/login.html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:9424d6c83228dfa0c2e18afaba1c8cc2e59b464c
 => => exporting config sha256:af3274cbb3ca1a0a88561f2fbf944099a6d887ec34
 => => exporting attestation manifest sha256:974c3ca0f96810f7e245ee0a1273
 => => exporting manifest list sha256:df5f26dcf6015a6af3efbab593cca26e782
 => => naming to docker.io/library/registration:1.0
 => => unpacking to docker.io/library/registration:1.0
```



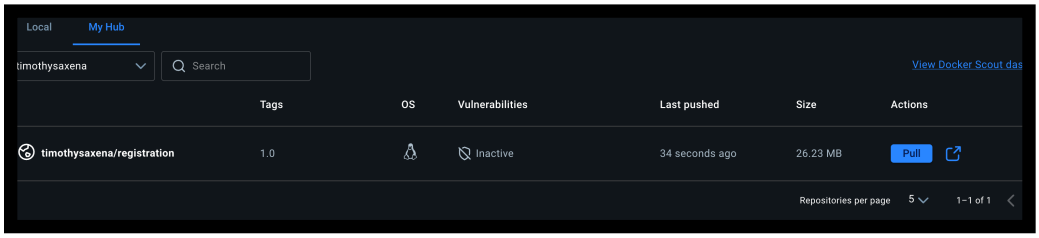


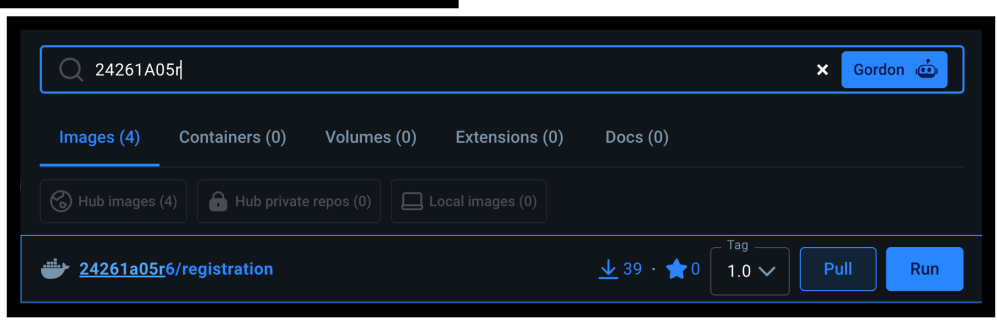
6

```
→  registration-app git:(main) ✗ docker tag registration:1.0 timothysaxena/registration:1.0
→  registration-app git:(main) ✗ docker login
Authenticating with existing credentials... [Username: timothysaxena]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
→  registration-app git:(main) ✗ docker push timothysaxena/registration:1.0
The push refers to repository [docker.io/timothysaxena/registration]
44136fa355b3: Pushed
858f76146c12: Pushed
a9986cd6f37d: Pushed
9894cf904239: Pushed
766d3311d5df: Pushed
ccea4a153c59: Pushed
943c389642fa: Pushed
bc862a9910f5: Pushed
d766d06974fa: Pushed
1efec2485f45: Pushed
ba1aad77d78d: Pushed
0bd71c05a58c: Pushed
1.0: digest: sha256:df5f26dcf6015a6af3efbab593cca26e782aa92df65de57b155e6abc290e75f7 size: 856
```





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